

Yes — this paper should become a **core research constraint** for Bundle 3.

The updated principle is:

A nice final pattern is not enough.

The system must produce mathematical models that generate falsifiable experimental predictions, perturbation designs, and mechanism-discrimination evidence.

Hiscock and Megason's key warning is that molecular, cellular, and mechanical mechanisms can all generate qualitatively similar periodic final patterns, so matching the final pattern is not sufficient evidence for the right mechanism. Instead, mathematical models should guide experiments that distinguish mechanisms by dynamics, perturbation response, parameter constraints, and interaction functions.

Updated Bundle 3 — NCA-ART-DSL mechanism discovery stack

Scrum-master view

Bundle objective: prepare the executable research loop for discovering, abstracting, and testing 5-node GRN patterning mechanisms.

This bundle is no longer just:

generate pattern -> score pattern

It is:

generate candidate mechanism
simulate dynamics
learn NCA cell-rule behavior
discover ART2/ARTMAP prototypes and transitions



map behavior back to DSL

ask: what experimental perturbation would distinguish this mechanism?



The core research loop becomes:

DSL candidate GRN

- > PDE/ODE simulator
- > NCA cell update model
- > ART2 prototype discovery
- > ARTMAP transition learning
- > prototype-to-DSL inverse mapping
- > mechanism-discrimination analysis
- > experiment-design proposal

This matches your existing architecture: simulator as the authoritative forward model, NCA as the learned/discrete cell update layer, ART as the prototype/transition discovery layer, DSL as the symbolic model layer, and verification through stability, sweeps, Bayesian/ABC, robustness, and re-simulation.

Updated user story

As the Research Scientist,
I want the NCA-ART-GRN repo to discover and test candidate 5-node GRN mechanisms,
so that I can move beyond visually matching Turing-like patterns and instead produce
symbolic models, NCA rules, ART2/ARTMAP transition evidence, and mathematical
experimental designs that help distinguish the real mechanism.

Updated concretizations / managed steps

```
prepare_dsl_candidate_runtime  
prepare_mechanism_hypothesis_runtime  
prepare_pde_ode_simulation_runtime  
prepare_nca_cell_runtime  
prepare_pde_ode_to_nca_dataset  
  
prepare_art2_discovery_runtime  
prepare_artmap_transition_runtime  
prepare_prototype_store  
prepare_transition_graph_store  
  
prepare_pattern_dynamics_metrics  
prepare_interaction_function_inference_runtime  
prepare_perturbation_design_runtime  
prepare_prototype_to_dsl_runtime  
prepare_mechanism_discrimination_report  
  
prepare_nca_art_smoke_configs  
check_nca_art_pipeline_inputs  
run_nca_art_local_smoke
```

New or changed steps are:

```
prepare_mechanism_hypothesis_runtime  
prepare_pattern_dynamics_metrics  
prepare_interaction_function_inference_runtime  
prepare_perturbation_design_runtime  
prepare_mechanism_discrimination_report
```

These are needed because the paper changes your success criteria.

What each step should do

`prepare_dsl_candidate_runtime`

Creates the symbolic candidate language.

The DSL must encode:

```
5-node GRN topology
activation/repression signs
interaction matrix
reaction parameters
diffusion parameters
motif provenance
known 2/3/4-node motif embedding
candidate constraints
experimental observables
perturbable parameters
```

This is primary, not optional. ART cannot replace the DSL because the DSL is what lets you express human-readable mechanisms, constraints, matrix dynamics, Turing checks, parameter scaling, and experiment design. The earlier research notes explicitly separate ART discovery from symbolic DSL analysis and verification.

`prepare_mechanism_hypothesis_runtime`

New step.

Creates a schema for mechanism hypotheses.

Path examples:

```
src/nca_art_grn/mechanisms/  
  hypothesis.py  
  classes.py  
  perturbations.py  
  observables.py  
  experimental_design.py  
  
configs/mechanisms/  
  molecular_reaction_diffusion.yaml  
  cellular_contact_or_motility.yaml  
  mechanical_or_boundary.yaml  
  grn_5node_molecular_default.yaml
```

Even if your current system is focused on molecular GRN reaction-diffusion, the schema should force every candidate to state:

```
mechanism_class  
  expected local activation source  
  expected long-range inhibition source  
  diffusion or transport assumptions  
  pattern-spacing prediction  
  dynamics prediction  
  perturbation predictions  
  required measurable parameters
```

Why: Hiscock and Megason argue that the final pattern cannot distinguish mechanisms; your DSL candidates therefore need hypotheses and predicted tests attached, not only pattern images.

`prepare_pde_ode_simulation_runtime`

Creates the authoritative forward simulator.

This remains the ground-truth generator:

```
5-node GRN ODE per cell
diffusion across grid
initial conditions
boundary conditions
perturbations
noise / heterogeneity
time-course output
```

But the simulator must now save **dynamics**, not just final states.

Required outputs:

```
full or sampled time series
pattern wavelength over time
growth rate of modes
candidate metadata
perturbation metadata
boundary condition metadata
diffusion metadata
```

Change: do not only save "final pattern score." Save enough to test mechanism.

`prepare_nca_cell_runtime`

Creates the NCA cell-update runtime.

NCA is integral. It should be treated as the learned/discrete local rule that helps answer:

```
Can the local update behavior implied by the PDE/ODE mechanism be learned,
```



The NCA should support:

```
neighborhood -> next center state  
multi-channel 5-node state  
rollout comparison with PDE/ODE  
perturbation replay  
local-rule inspection  
prototype-conditioned rule testing
```

Your previous research plan frames NCA as a per-cell update function trained from simulator trajectories or used to test ART-derived local rules.

`prepare_pde_ode_to_nca_dataset`

Creates the simulator-to-NCA dataset contract.

Dataset rows should include:

```
candidate_id  
mechanism_hypothesis_id  
time_t  
time_t_plus_1  
center_state_t  
neighborhood_state_t  
center_state_next  
diffusion_parameters  
reaction_parameters  
boundary_conditions  
perturbation_id  
pattern_metrics_at_t
```

Change: include `mechanism_hypothesis_id` and `perturbation_id`, because later you need to know which mechanism produced which local rule and which perturbations tested it.

`prepare_art2_discovery_runtime`

Creates ART2-centered continuous prototype discovery.

ART2 is primary because your data are continuous state vectors from PDE/ODE and NCA, not just discrete labels.

ART2 should cluster:

- center-state vectors
- center + neighborhood vectors
- NCA hidden/local update vectors
- perturbed-state vectors
- time-window vectors

Outputs:

- prototype vectors
- prototype assignment maps
- prototype support counts
- prototype birth/update events
- prototype stability statistics
- representative local patches

Change: ART2 prototypes must be tied to mechanism and perturbation metadata, not only pattern classes.

`prepare_artmap_transition_runtime`

Creates ARTMAP/regression transition learning.

ARTMAP is primary because this project is about mechanisms and local transitions, not only clusters.

ARTMAP should learn mappings such as:

```
neighborhood state at t -> center state at t+1  
prototype at t + context -> prototype at t+1  
perturbed local state -> recovery trajectory  
DSL candidate parameters -> expected prototype transition family
```

Why: ARTMAP gives you a bridge between observed simulator/NCA dynamics and local rule behavior. That is directly useful for distinguishing mechanisms because mechanism differences often appear in **dynamics**, not final pattern appearance.

prepare_prototype_store

Creates persistent prototype artifacts.

Add required fields:

```
prototype_id  
method = ART2 / FuzzyART / TopoART / DualVigilance  
mechanism_hypothesis_id  
candidate_id  
input_schema  
prototype_vector  
support_count  
source_runs  
perturbation_contexts
```

prepare_transition_graph_store

Creates persistent transition graph artifacts.

Add required fields:

```
source_prototype
target_prototype
frequency
mechanism_hypothesis_id
candidate_id
perturbation_id
time_delta
context_features
transition_score
recovery_or_failure_flag
```

Change: transitions should support mechanism testing: how does a local state respond after perturbation, boundary change, diffusion change, or initial-condition change?

prepare_pattern_dynamics_metrics

New step.

Creates metrics that look at formation dynamics, not just final pattern.

Path examples:

```
src/nca_art_grn/analysis/pattern_dynamics.py
src/nca_art_grn/analysis/wavelength.py
```

```
src/nca_art_grn/analysis/dispersion.py  
src/nca_art_grn/analysis/mode_growth.py  
src/nca_art_grn/analysis/boundary_effects.py
```

Metrics:

```
pattern wavelength over time  
dominant Fourier mode  
mode growth rate  
time to pattern emergence  
stripe/spot/labyrinth classification  
boundary sensitivity  
initial-condition sensitivity  
pattern recovery after perturbation
```

Why: Hiscock and Megason stress that final pattern appearance is not enough; pattern formation dynamics and perturbation responses are more informative.

`prepare_interaction_function_inference_runtime`

New step.

Implements the $\kappa(x)$ -style interaction inference idea.

Path examples:

```
src/nca_art_grn/analysis/interaction_function.py  
src/nca_art_grn/analysis/k_estimation.py  
src/nca_art_grn/analysis/local_response.py  
configs/analysis/interaction_function_smoke.yaml
```

Purpose:

```
estimate how local states influence nearby states over distance  
compare smooth molecular-like interactions vs sharp cellular-like interactions  
compare long-range/boundary/mechanical-like effects
```

Even if your first implementation is molecular GRN/PDE focused, this is valuable because it prevents the system from claiming “Turing mechanism” based only on final pattern similarity.

Hiscock and Megason describe estimating interaction functions from pattern dynamics as a way to distinguish molecular, cellular, and mechanical interaction classes.

`prepare_perturbation_design_runtime`

New step.

Creates perturbation configs that are designed to distinguish mechanisms.

Path examples:

```
src/nca_art_grn/analysis/perturbation_design.py  
src/nca_art_grn/analysis/scaling_tests.py  
src/nca_art_grn/analysis/experimental_predictions.py  
  
configs/perturbations/  
  diffusion_scaling.yaml  
  degradation_half_life.yaml  
  boundary_condition_shift.yaml  
  initial_condition_bias.yaml  
  local_ablation.yaml  
  noise_heterogeneity.yaml
```

The system should generate predictions like:

If molecular diffusion controls spacing:
increasing diffusion should scale pattern wavelength predictably.

If boundary conditions drive orientation:
changing boundary geometry should alter orientation or spot/stripe selection.

If initial conditions dominate:
changing the seed stripe should bias orientation.

If local rule is robust:
NCA and PDE/ODE should recover similar pattern after perturbation.

Why: Hiscock and Megason emphasize perturbations and scaling laws as stronger tests than visual similarity.

prepare_prototype_to_dsl_runtime

Updated.

This still maps ART/ARTMAP output back to DSL, but now the DSL candidate must include mechanism-test metadata.

Output DSL artifact should include:

candidate topology
parameters
diffusion values
motif core
prototype sources
transition sources
Turing/stability report placeholder

perturbation predictions

Your existing architecture already maps prototypes and transition data into symbolic GRN models using optimization, sparse regression, sign constraints, and motif reduction.

Change: prototype-to-DSL mapping must not end at “this produces a pattern.” It must produce “this mechanism predicts these perturbation outcomes.”

`prepare_mechanism_discrimination_report`

New step.

Creates a report template for each candidate.

Path:

`/workspace/artifacts/nca-art-grn/mechanism_reports/`

Report fields:

```
candidate_id
mechanism_hypothesis
final_pattern_summary
formation_dynamics_summary
NCA agreement summary
ART2 prototype summary
ARTMAP transition summary
DSL mapping summary
perturbation predictions
parameter constraints
```

This is the most important update.

The report must explicitly answer:

What would I test experimentally to know this is the right mechanism?

That is the paper's core lesson.

`prepare_nca_art_smoke_configs`

Update the smoke configs to include one mechanism-discrimination mini-test.

Add:

```
configs/experiments/smoke_mechanism_discrimination.yaml
configs/perturbations/smoke_diffusion_scaling.yaml
configs/analysis/smoke_pattern_dynamics.yaml
configs/analysis/smoke_interaction_function.yaml
```

Smoke should remain tiny:

```
1 candidate
8x8 or 16x16 grid
few time steps
one diffusion perturbation
one ART2 pass
one ARTMAP transition pass
one NCA mini rollout
one mechanism report
```

check_nca_art_pipeline_inputs

Updated readiness check should report:

```
DSL runtime ready
mechanism hypothesis configs ready
PDE/ODE simulator configs ready
NCA configs ready
ART2 configs ready
ARTMAP configs ready
prototype store ready
transition graph store ready
pattern dynamics metrics ready
perturbation configs ready
mechanism report path ready
```

run_nca_art_local_smoke

Updated explicit smoke execution.

It should run:

```
load one DSL candidate
attach mechanism hypothesis
run tiny PDE/ODE simulation
capture trajectory
train/test tiny NCA rollout
run ART2 prototype discovery
run ARTMAP transition mapping
calculate basic dynamics metrics
run one perturbation comparison
```


attempt prototype-to-DSL mapping
write mechanism-discrimination report

Expected output:

```
/workspace/runs/nca-art-grn/smoke/<timestamp>/  
metadata.json  
candidate.dsl.json  
simulator_summary.json  
nca_summary.json  
art2_prototypes.json  
artmap_transitions.json  
pattern_dynamics.json  
perturbation_summary.json  
mechanism_report.md
```

Updated acceptance criteria

The following should be valid:

```
sudo config --target researchscientist bootstrap step prepare_dsl_candidate_runtime  
sudo config --target researchscientist bootstrap step prepare_mechanism_hypothesis_runtime  
sudo config --target researchscientist bootstrap step prepare_pde_ode_simulation_runtime  
sudo config --target researchscientist bootstrap step prepare_nca_cell_runtime  
sudo config --target researchscientist bootstrap step prepare_pde_ode_to_nca_dataset  
sudo config --target researchscientist bootstrap step prepare_art2_discovery_runtime  
sudo config --target researchscientist bootstrap step prepare_artmap_transition_runtime  
sudo config --target researchscientist bootstrap step prepare_pattern_dynamics_metrics  
sudo config --target researchscientist bootstrap step prepare_interaction_function_inference_runtime  
sudo config --target researchscientist bootstrap step prepare_perturbation_design_runtime  
sudo config --target researchscientist bootstrap step prepare_prototype_to_dsl_runtime
```

```
sudo config --target researchscientist bootstrap step prepare_mechanism_discrimination_report
```

Explicit execution:

```
sudo config --target researchscientist bootstrap step run_nca_art_local_smoke
```

Prepare/check steps must:

```
not run large simulations
not train full NCA models
not launch Runpod
not run parameter campaigns
not overwrite research code
not overwrite existing artifacts
not claim final biological mechanism from pattern appearance alone
not build paper output
```

Smoke step must:

```
use tiny local config
record mechanism hypothesis
record dynamics metrics
record perturbation metadata
record ART2 and ARTMAP settings
write inspectable mechanism report
clearly label outputs as smoke, not discovery result
```

Updated tests

Registry tests

```
config bootstrap steps | grep prepare_mechanism_hypothesis_runtime
config bootstrap steps | grep prepare_pattern_dynamics_metrics
config bootstrap steps | grep prepare_interaction_function_inference_runtime
config bootstrap steps | grep prepare_perturbation_design_runtime
config bootstrap steps | grep prepare_mechanism_discrimination_report
```

Structure tests

```
test -d /workspace/repos/nca-art-grn/src/nca_art_grn/mechanisms
test -d /workspace/repos/nca-art-grn/src/nca_art_grn/analysis
test -d /workspace/repos/nca-art-grn/configs/mechanisms
test -d /workspace/repos/nca-art-grn/configs/perturbations
test -d /workspace/artifacts/nca-art-grn/mechanism_reports
```

Config tests

```
test -f /workspace/repos/nca-art-grn/configs/mechanisms/grn_5node_molecular_default.yaml
test -f /workspace/repos/nca-art-grn/configs/perturbations/diffusion_scaling.yaml
test -f /workspace/repos/nca-art-grn/configs/analysis/smoke_pattern_dynamics.yaml
test -f /workspace/repos/nca-art-grn/configs/analysis/smoke_interaction_function.yaml
```

Smoke output tests

```
sudo config --target researchscientist bootstrap step run_nca_art_local_smoke

find /workspace/runs/nca-art-grn/smoke -name mechanism_report.md | tail -n 1
find /workspace/runs/nca-art-grn/smoke -name pattern_dynamics.json | tail -n 1
find /workspace/runs/nca-art-grn/smoke -name perturbation_summary.json | tail -n 1
find /workspace/runs/nca-art-grn/smoke -name art2_prototypes.json | tail -n 1
find /workspace/runs/nca-art-grn/smoke -name artmap_transitions.json | tail -n 1
```

Scientific guardrail test

The smoke report must contain these headings:

```
Final pattern is not sufficient evidence
Mechanism hypothesis
Dynamics evidence
Perturbation prediction
Experimental design suggestion
Falsification criterion
```

Updated Bundle 3 summary

Bundle 3 now has the correct scientific core:

```
The goal is not to find pretty Turing-like images.

The goal is to find candidate GRN mechanisms that:
  produce patterning dynamics,
  can be represented in DSL,
  can be approximated/tested by NCA,
  can be summarized by ART2 and ARTMAP,
  can survive perturbation/robustness checks,
  and can suggest experiments that distinguish the mechanism.
```

This makes Hiscock and Megason part of your research logic:

```
Pattern match = weak evidence.
Dynamics + perturbation response + parameter constraints + experimental design = strong evidence.
```

So Bundle 3 becomes the foundation for a serious MRes/research platform, not just a pattern generator.

you left out the what does this do for the research, and also its not clear if this is a config, agentfield etc step and if it is part of that managed role-python-etc platform thing if not what and how does this steplink into the whole system?

Nachgedacht für 5s

You are right. I mixed two levels together.

For **Bundle 3**, most steps are **not Agentfield steps yet**. They are mostly **config-managed bootstrap steps** that prepare the research repo and its runtime contracts. The actual scientific code lives inside the `nca-art-grn` repo. Later, Agentfield will call these repo entrypoints, but Agentfield does not own them yet.

Corrected Bundle 3 framing

Where this sits in the whole system

```
config
  owns setup:
    users, roles, envs, packages, folders, configs, smoke commands

nca-art-grn repo
  owns research code:
    DSL, PDE/ODE simulator, NCA, ART2, ARTMAP, mapping, analysis

/workspace/data, /workspace/runs, /workspace/artifacts
  own large data, run outputs, prototypes, reports, figures
```

```
Agentfield later
  owns orchestration:
    "run this experiment", track status, compare runs, call repo CLI
```

```
Paperclip later
  owns human dashboard:
    launch/review runs, inspect status/artifacts/results
```

So Bundle 3 steps are mostly:

```
config-managed preparation steps
  that create/verify repo files, configs, folders, and tiny smoke entrypoints
```

They are **not** full experiments yet, and **not** Agentfield CRDs yet.

Bundle 3 — NCA-ART-DSL mechanism discovery stack

Scrum-master view

Bundle objective: prepare the `nca-art-grn` repo so that the Research Scientist can run the first real local mechanism-discovery loop.

The research principle from Hiscock and Megason is now a hard requirement:

```
Final pattern match is weak evidence.
Mechanism-discriminating predictions are strong evidence.
```

They argue that molecular, cellular, and mechanical mechanisms can produce qualitatively similar final periodic patterns; therefore, mathematical models should guide experiments using dynamics, perturbations, parameter constraints, and interaction functions, not just final pattern similarity.

User story

As the Research Scientist,
I want config to prepare the NCA-ART-GRN research engine,
so that I can generate DSL-defined GRN mechanisms, simulate them, train/test NCA
local rules, discover ART2/ARTMAP prototype transitions, and produce mechanism
reports that propose experiments to distinguish the right patterning mechanism.

Step type legend

Use this to keep the platform clean:

CONFIG STEP

A managed bootstrap/check/smoke step invoked by config.

REPO CODE

Python modules, configs, tests, and scripts inside nca-art-grn.

RESEARCH OUTPUT

Data, runs, artifacts, prototypes, transition graphs, reports.

AGENTFIELD LATER

Not implemented here; later calls the repo CLI/modules.

Concretizations with role, ownership, and research meaning

prepare_dsl_candidate_runtime

Type: CONFIG STEP

Owner: config prepares it; researchscientist uses it.

Creates/validates REPO CODE:

```
src/nca_art_grn/dsl/  
configs/dsl/
```

What this does for the research:

This makes the DSL the formal language for your candidate mechanisms. It stores 5-node GRN topology, activation/repression signs, parameters, diffusion assumptions, motif provenance, perturbable parameters, and later inverse-mapped candidates from ART/ARTMAP.

How it links into the platform:

```
config creates DSL skeleton/configs  
research code uses DSL to generate candidates  
runs write candidate DSL snapshots to /workspace/runs/nca-art-grn  
Agentfield later submits "candidate DSL + experiment config" as a run input  
Paperclip later displays candidate DSL summary and mechanism report
```

prepare_mechanism_hypothesis_runtime

Type: CONFIG STEP

Owner: config prepares mechanism schema/configs.

Creates/validates REPO CODE:


```
src/nca_art_grn/mechanisms/
configs/mechanisms/
```

What this does for the research:

This step forces every candidate to declare what mechanism it claims to represent. It is where Hiscock/Megason becomes operational: a candidate must state its local activation, long-range inhibition, expected dynamics, perturbation predictions, and falsification criteria.

How it links into the platform:

```
DSL candidate gets mechanism_hypothesis_id
simulator and NCA outputs carry that ID
ART2/ARTMAP prototypes keep that ID
mechanism report summarizes whether evidence supports that hypothesis
Agentfield later groups runs by mechanism_hypothesis_id
```

prepare_pde_ode_simulation_runtime

Type: CONFIG STEP

Owner: config prepares skeleton/configs; repo owns simulator implementation.

Creates/validates REPO CODE:

```
src/nca_art_grn/simulator/
configs/experiments/pde_ode_baseline.yaml
```

What this does for the research:

This creates the authoritative forward model. It should simulate 5-node GRN reaction-diffusion dynamics and save time-series evidence, not just a final image.

How it links into the platform:

```
DSL candidate -> simulator config  
simulator -> /workspace/runs/nca-art-grn/<run_id>/trajectories  
simulator -> NCA dataset writer  
simulator -> ART2/ARTMAP input  
simulator -> mechanism report dynamics evidence
```

prepare_nca_cell_runtime

Type: CONFIG STEP

Owner: config prepares NCA module/config placeholders; repo owns implementation.

Creates/validates REPO CODE:

```
src/nca_art_grn/nca/  
configs/nca/
```

What this does for the research:

NCA is integral. It is the learned/discrete local cell-rule layer. It tests whether local neighborhood dynamics from the PDE/ODE mechanism can be learned, replayed, perturbed, compressed, and compared against ART/ARTMAP transitions.

How it links into the platform:

```
PDE/ODE trajectories -> NCA training/evaluation samples  
NCA rollout -> comparison against simulator  
NCA local states -> ART2 prototype discovery  
NCA rule behavior -> mechanism report  
Agentfield later can choose simulator-only, NCA-only, or simulator+NCA experiment modes
```

prepare_pde_ode_to_nca_dataset

Type: CONFIG STEP

Owner: config prepares dataset contracts.

Creates/validates REPO CODE + RESEARCH OUTPUT contract:

```
src/nca_art_grn/capture/  
configs/data/
```

What this does for the research:

This makes the simulator output trainable for NCA and analyzable by ART/ARTMAP. It defines rows like neighborhood at t , center state at $t+1$, mechanism hypothesis ID, perturbation ID, and pattern metrics.

How it links into the platform:

```
simulator output -> /workspace/data/nca-art-grn/datasets/  
NCA training reads that dataset  
ART2/ARTMAP can read same state-transition samples  
Agentfield later registers dataset IDs as experiment inputs
```

prepare_art2_discovery_runtime

Type: CONFIG STEP

Owner: config prepares ART2 module/config placeholders.

Creates/validates REPO CODE:

```
src/nca_art_grn/art/art2.py  
src/nca_art_grn/art/prototypes.py
```

config/art/art2_baseline.yaml

What this does for the research:

ART2 is the continuous-state prototype discovery layer. It finds recurring local concentration/state patterns from simulator and NCA trajectories. This gives you a vocabulary of local states rather than relying only on whole-image pattern scores.

How it links into the platform:

```
simulator/NCA state vectors -> ART2
ART2 -> /workspace/artifacts/nca-art-grn/prototypes/art2
prototype IDs -> ARTMAP transition learning
prototype summaries -> mechanism report
Paperclip later can show prototypes as reviewable artifacts
```

prepare_artmap_transition_runtime

Type: CONFIG STEP

Owner: config prepares ARTMAP module/config placeholders.

Creates/validates REPO CODE:

```
src/nca_art_grn/art/artmap.py
src/nca_art_grn/art/transition_model.py
configs/artmap/
```

What this does for the research:

ARTMAP is the transition-learning layer. It connects local state/context at t to prototype or state at $t+1$. This is central because mechanism differences often appear in **dynamics**, not final pattern shape.

How it links into the platform:

```
ART2 prototypes + transition samples -> ARTMAP  
ARTMAP -> transition graph artifacts  
transition graph -> prototype-to-DSL mapping  
transition evidence -> mechanism report  
Agentfield later compares transition models across candidates
```

prepare_pattern_dynamics_metrics

Type: CONFIG STEP

Owner: config prepares analysis modules/configs.

Creates/validates REPO CODE:

```
src/nca_art_grn/analysis/pattern_dynamics.py  
src/nca_art_grn/analysis/wavelength.py  
src/nca_art_grn/analysis/dispersion.py  
src/nca_art_grn/analysis/mode_growth.py
```

What this does for the research:

This prevents “pretty final pattern” from being the criterion. It measures wavelength over time, dominant Fourier modes, mode growth, time-to-pattern, boundary sensitivity, recovery after perturbation, and pattern dynamics.

How it links into the platform:

```
simulator/NCA runs -> dynamics metrics  
dynamics metrics -> mechanism discrimination report  
Bundle 4 later uses metrics for search/optimization objectives  
Agentfield later tracks these metrics as run status/results
```

prepare_interaction_function_inference_runtime

Type: CONFIG STEP

Owner: config prepares analysis runtime.

Creates/validates REPO CODE:

```
src/nca_art_grn/analysis/interaction_function.py  
src/nca_art_grn/analysis/k_estimation.py  
configs/analysis/interaction_function_smoke.yaml
```

What this does for the research:

This implements the Hiscock/Megason idea of estimating interaction functions from dynamics. It helps ask whether the observed behavior looks like smooth molecular diffusion, sharp cellular contact, or longer-range mechanical/boundary effects.

How it links into the platform:

```
trajectory dynamics -> K(x)-style estimate  
K(x) estimate -> mechanism evidence  
mechanism evidence -> report  
later Agentfield can compare K(x) estimates across hypotheses
```

prepare_perturbation_design_runtime

Type: CONFIG STEP

Owner: config prepares perturbation configs and analysis modules.

Creates/validates REPO CODE:

```
src/nca_art_grn/analysis/perturbation_design.py  
src/nca_art_grn/analysis/scaling_tests.py
```



configs/perturbations/



What this does for the research:

This turns mathematical models into experimental design suggestions. It defines perturbations such as diffusion scaling, degradation/half-life changes, boundary shifts, initial-condition bias, local ablation, and noise/heterogeneity.

How it links into the platform:

```
mechanism hypothesis -> perturbation plan
perturbation plan -> simulator/NCA rerun config
perturbation results -> mechanism report
Bundle 4 later uses perturbations in robustness/search
Agentfield later can schedule perturbation campaigns
```

prepare_prototype_store

Type: CONFIG STEP

Owner: config creates artifact directories/schema.

Creates RESEARCH OUTPUT paths:

/workspace/artifacts/nca-art-grn/prototypes/

What this does for the research:

Keeps ART2/ART prototype artifacts traceable to candidate ID, mechanism hypothesis, simulator/NCA run, timepoint, perturbation context, and support count.

How it links into the platform:

ART2 writes prototypes here
prototype-to-DSL reads from here
mechanism report cites these artifacts
Paperclip later displays these outputs

prepare_transition_graph_store

Type: CONFIG STEP

Owner: config creates artifact directories/schema.

Creates RESEARCH OUTPUT paths:

/workspace/artifacts/nca-art-grn/transition_graphs/

What this does for the research:

Stores the local behavior graph: which states become which states, under which candidate, perturbation, and mechanism hypothesis.

How it links into the platform:

ARTMAP writes transition graphs
prototype-to-DSL uses transition graphs
mechanism report summarizes transition evidence
Agentfield later compares transition graphs across runs

prepare_prototype_to_dsl_runtime

Type: CONFIG STEP

Owner: config prepares mapping modules/configs.

Creates/validates REPO CODE:

```
src/nca_art_grn/mapping/  
configs/mapping/
```

What this does for the research:

This closes the loop. It attempts to map ART2 prototypes and ARTMAP transitions back into symbolic DSL candidates with topology, signs, parameters, constraints, and perturbation predictions.

How it links into the platform:

```
ART2/ARTMAP artifacts -> prototype-to-DSL mapper  
mapper -> candidate DSL artifact  
candidate DSL -> resimulation  
candidate DSL -> paper/report material  
Agentfield later can branch new experiments from mapped candidates
```

```
prepare_mechanism_discrimination_report
```

Type: CONFIG STEP

Owner: config prepares report templates/paths.

Creates RESEARCH OUTPUT path:

```
/workspace/artifacts/nca-art-grn/mechanism_reports/
```

What this does for the research:

This is the scientific checkpoint. Every candidate report must answer:

What mechanism is being claimed?
Why is final pattern similarity insufficient?
What dynamics support it?
What perturbation would falsify it?
What experiment would distinguish it from alternatives?

How it links into the platform:

run outputs -> mechanism report
Bundle 10 later cites reports in paper
Agentfield later stores report path in experiment status
Paperclip later shows report for human review

prepare_nca_art_smoke_configs

Type: CONFIG STEP

Owner: config prepares tiny configs only.

Creates/validates REPO CONFIGS:

configs/experiments/smoke_nca_art_pipeline.yaml
configs/experiments/smoke_mechanism_discrimination.yaml
configs/perturbations/smoke_diffusion_scaling.yaml

What this does for the research:

Creates a safe tiny end-to-end test. It proves the pipeline is wired without pretending the result is a discovery.

How it links into the platform:

config can run one tiny smoke test
later Agentfield can reuse same config shape for real experiments

check_nca_art_pipeline_inputs

Type: CONFIG CHECK STEP

Owner: config .

Runs: no science, no training.

What this does for the research:

Confirms that all research pipeline pieces exist before wasting time on a broken run.

How it links into the platform:

operator can check readiness
Research Scientist can see missing pieces
Agentfield later can call equivalent readiness check before scheduling jobs

run_nca_art_local_smoke

Type: CONFIG SMOKE EXECUTION STEP

Owner: config invokes tiny repo command as researchscientist .

Runs: tiny local smoke only.

What this does for the research:

Proves the complete loop exists:

```
DSL candidate
-> simulator
-> NCA mini rollout
-> ART2 prototype discovery
-> ARTMAP transition mapping
-> dynamics metrics
-> perturbation mini-test
-> mechanism report
```

How it links into the platform:

```
today:
  config invokes repo smoke command

later:
  Agentfield invokes the same repo command with real experiment configs

later still:
  Paperclip starts/reviews the Agentfield run
```

Output:

```
/workspace/runs/nca-art-grn/smoke/<timestamp>/
```

Whole-system linkage

The pipeline should be designed like this:

```
config step
  prepares or checks files/configs/folders
```





```
repo CLI
  performs research actions

workspace paths
  store inputs, runs, artifacts

Agentfield later
  orchestrates repo CLI with experiment status

Paperclip later
  exposes Agentfield runs to user
```

Concrete later call shape:

```
config:
  sudo config --target researchscientist bootstrap step run_nca_art_local_smoke

repo:
  python -m nca_art_grn.cli.run --config configs/experiments/smoke_nca_art_pipeline.yaml

Agentfield later:
  GRNExperiment.spec.method = nca_art_pipeline
  GRNExperiment.spec.config = configs/experiments/...
  controller calls repo CLI
  status points to /workspace/runs/nca-art-grn/<run_id>

Paperclip later:
  shows run status, prototype artifacts, transition graphs, mechanism report
```

Updated summary

Bundle 3 is a **config-managed research-engine preparation bundle**.

It is not Agentfield yet.

```
config prepares the engine  
nca-art-grn contains the engine  
/workspace stores the evidence  
Agentfield later orchestrates the engine  
Paperclip later makes the engine visible
```

And scientifically:

```
DSL defines candidate mechanisms.  
PDE/ODE generates dynamics.  
NCA learns/tests local update rules.  
ART2 discovers continuous prototypes.  
ARTMAP learns transitions.  
Perturbation/dynamics analysis tests mechanism claims.  
Prototype-to-DSL closes the explanation loop.  
Mechanism reports prevent "pretty pattern" from becoming false evidence.
```